

Quality assurance vs. quality control explained: 5 key differences | by Qualio | Medium

By Qualio

Quality assurance vs. quality control explained: 5 key differences In the healthcare and life science industries, quality processes like quality assurance (QA) and quality control (QC) are required by the FDA and the International Organization for Standardization's ISO 9000. If QA and QC aren't followed, you could end up distributing an end product that not only doesn't meet standards, but could also have life-threatening consequences. Both QA and QC processes are important to quality standards, but so is knowing the difference between quality assurance and quality control. They're often used interchangeably, but they are two different processes taking place at different times. Quality assurance and quality control both play vital and distinct roles in the health of life sciences companies and are big parts of quality management. Understanding those roles can help your organization master each to deliver the best products possible. What is quality assurance? Quality assurance is a part of your quality management plan. QA encompasses all of the activities within your plan that ensure your product is ready to go through the manufacturing process (or software development process if you're creating software). In other words, you're verifying that the quality requirements you planned for will be fulfilled as the products are made. Quality control is the inspection phase of quality assurance. It's a series of test procedures used to verify that a product is safe and effective after mass production. Both QA and QC are necessary. You can't "test quality into" a product by relying on QC alone. There are different methodologies for both QA and QC, like Agile, Six Sigma, and other project management or process improvement strategies. They're designed to save time while managing your quality and project development. Using these can help, but you'll likely need a quality management system that follows ISO 9001:2015 regulations to effectively manage the entire quality planning process. Quality assurance vs. quality control: 5 differences To further understand the differences between quality assurance and quality control, you need to get a picture of how the two processes work together to boost your organization's quality improvement and help reduce corrective actions.

1. Proactive (QA) vs. reactive (QC) Effective quality assurance is proactive. It aims to prevent defects before they occur through process design. QC is reactive and exists to identify defects in the quality of products after they have happened. QA involves the design of processes, such as documenting standard operating procedures (SOPs) according to ISO 9000 standards. A safe, effective product should be the result every time processes are followed. QC involves the testing of products to ensure they meet standards for safety and efficacy. If QC testing uncovers quality issues, it should result in reactive steps to prevent an unsafe product from being shipped and distributed. Ideally, QC issues should also spark a QA review. Non-conforming test results should result in corrective and preventive action (CAPA) investigation to determine the root cause of quality issues and update processes to prevent the problem from happening in the future.

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2. Process (QA) vs. product (QC) QA is process-oriented, and it focuses on preventing quality issues. QC is product-oriented and focused on identifying quality issues in manufactured products that could affect customer satisfaction. Another way to understand this distinction is actions vs. results. QA involves the actions which create the product, while QC is focused on the resulting product. Several examples of each type of activity are detailed below. QA processes: Documentation Audits Supplier management Personnel training Change control Investigation procedures QC procedures: Batch inspection Product sampling Validation testing Laboratory testing Software testing
3. System (QA) vs. parts (QC) Quality assurance control systems are the methods and procedures which are used to safeguard quality standards. Quality control systems measure parts, including the

outputs of the system. QC efforts may also be focused on parts used to create the final product, such as raw materials from a supplier. The QA system for quality management may dictate various activities to make sure inputs are consistently safe and effective, such as auditing suppliers and batch sampling raw materials. RELATED READING: The 5 Best Medical Device Quality Assurance Training Options

4. Creation (QA) vs. verification (QC)

The result of QA activities is a roadmap for creating high-quality products. It involves defining standards for product design, manufacture, packaging, distribution, marketing, and sales. QC involves verification of products post-manufacture and before distribution, or confirming safety and efficacy.

5. Entire team (QA) vs. dedicated personnel (QC)

Quality assurance activities involve the entire team. Every member of a life sciences organization is responsible for QA activities by following SOPs. While the quality management system (QMS) is generally the responsibility of the quality unit and the leadership team, QA activities involve standards for training, documentation, and review across the workforce. QC is generally the responsibility of certain personnel within the organization whose duties include following SOPs for product testing. QC staff follow SOPs for quality control and document their findings based on standardized procedures for product testing and process validation. The role of an eQMS in quality assurance and quality control

Neither QA or QC are optional. Similarly, it's impossible to say whether QA or QC is more valuable. QA involves creating standards and processes to create a safe, effective process. QC activities validate the product. Life sciences organizations should "close the loop" on quality management processes by using QC to inform QA. When non-conformances are discovered, it's important to launch an investigation to determine where the breakdown occurred in QA processes and create stronger standards. QC and QA are better together, and they're best when both are incorporated into an enterprise quality management system (eQMS) for end-to-end quality processes. A single, cloud-based QMS for life sciences can integrate processes for assuring and controlling quality through document control and training, CAPA management and complaint handling throughout the product life cycle. Qualio's eQMS is built specifically for start-up and scale-up life science companies. If your company needs a scalable, simple QMS platform, we'd love to show you what Qualio can do for your quality assurance and quality control with a personalized demo.